

\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. ReportExample2.
AUTHOR. Michael Coughlan.

Click to view the program
commentary

ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
 SELECT SalesFile ASSIGN TO "GBSALES.DAT"
 ORGANIZATION IS LINE SEQUENTIAL.
 SELECT PrintFile ASSIGN TO "SALESREPORT2.LPT".

DATA DIVISION.
FILE SECTION.
FD SalesFile.
01 SalesRecord.
 88 EndOfFile VALUE HIGH-VALUES.
 02 CityCode PIC 9.
 02 SalesPersonNum PIC 9.
 02 ValueOfSale PIC 9(4)V99.

FD PrintFile
 REPORT IS SalesReport.

WORKING-STORAGE SECTION.
01 NameTable.
 02 TableValues.
 03 FILLER PIC X(18) VALUE "Dublin Belfast ".
 03 FILLER PIC X(18) VALUE "Cork Galway ".
 03 FILLER PIC X(18) VALUE "Sligo Waterford".
 03 FILLER PIC X(9) VALUE "Limerick".
 02 FILLER REDEFINES TableValues.
 03 CityName PIC X(9) OCCURS 7 TIMES.

REPORT SECTION.
RD SalesReport
 CONTROLS ARE SalesPersonNum
 PAGE LIMIT IS 66
 HEADING 1
 FIRST DETAIL 6
 LAST DETAIL 42
 FOOTING 52.

\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. ReportExample2.
AUTHOR. Michael Coughlan.

ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.

SELECT SalesFile ASSIGN TO "GBSALES.DAT"
ORGANIZATION IS LINE SEQUENTIAL.
SELECT PrintFile ASSIGN TO "SALESREPORT2.LPT".

DATA DIVISION.
FILE SECTION.
FD SalesFile.
01 SalesRecord.
88 EndOfFile VALUE HIGH-VALUES.
02 CityCode PIC 9.
02 SalesPersonNum PIC 9.
02 ValueOfSale PIC 9(4)V99.

FD PrintFile
REPORT IS SalesReport.

WORKING-STORAGE SECTION
01 NameTable.
02 TableValues.
03 FILLER PIC X(18) VALUE "Dublin Belfast".
03 FILLER PIC X(18) VALUE "Cork Galway".
03 FILLER PIC X(18) VALUE "Sligo Waterford".
03 FILLER PIC X(9) VALUE "Limerick".
02 FILLER REDEFINES TableValues.
03 CityName PIC X(9) OCCURS 7 TIMES.

REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.

The FD entry of an ordinary print file would be followed by a record description of the print line such as -

FD PrintFile.
01 PrintLine PIC X(80).

```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. ReportExample2.
AUTHOR. Michael Coughlan.
```

```
ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
```

```
SELECT SalesFile ASSIGN TO "GBSALES.DAT"
      ORGANIZATION IS LINE SEQUENTIAL.
SELECT PrintFile ASSIGN TO "SALESREPORT2.LPT".
```

```
DATA DIVISION.
FILE SECTION.
FD SalesFile.
01 SalesRecord.
   88 EndOfFile VALUE HIGH-VALUES.
   02 CityCode PIC 9.
   02 SalesPersonNum PIC 9.
   02 ValueOfSale PIC 9(4)V99.
```

```
FD PrintFile
REPORT IS SalesReport.
```

```
WORKING-STORAGE SECTION.
```

```
01 NameTable.
   02 TableValues.
      03 FILLER PIC X(18) VALUE "Dublin Belfast ".
      03 FILLER PIC X(18) VALUE "Cork Galway ".
      03 FILLER PIC X(18) VALUE "Sligo Waterford".
      03 FILLER PIC X(9) VALUE "Limerick".
   02 FILLER REDEFINES TableValues.
      03 CityName PIC X(9) OCCURS 7 TIMES.
```

```
REPORT SECTION.
```

```
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.
```

But in the Report Writer
the FD entry contains a
clause linking the file
name with a Report
described in the **REPORT
SECTION**.

```

REPORT SECTION.
RD   SalesReport
     CONTROLS ARE SalesPersonNum
     PAGE LIMIT IS 66
     HEADING 1
     FIRST DETAIL 6
     LAST DETAIL 42
     FOOTING 52.

```

In this case it is a PAGE
HEADING group.

Note that we do not need to
give the group record a name if
we don't refer to it later.

```

01  TYPE IS PAGE HEADING.
02  LINE 1.
03  COLUMN 12          PIC X(32)
                           VALUE "An example COBOL Report Program".

02  LINE 2.
03  COLUMN 6           PIC X(17)
                           VALUE "Bible Salesperson".
03  COLUMN 23          PIC X(26)
                           VALUE " - Sales and Salary Report".

02  LINE 4.
03  COLUMN 2           PIC X(4)  VALUE "City".
03  COLUMN 12          PIC X(11) VALUE "Salesperson".
03  COLUMN 28          PIC X(4)  VALUE "Sale".

02  LINE 5.
03  COLUMN 2           PIC X(4)  VALUE "Name".
03  COLUMN 13          PIC X(6)  VALUE "Number".
03  COLUMN 28          PIC X(5)  VALUE "Value".

01  DetailLine TYPE IS DETAIL.
02  LINE IS PLUS 1.
03  COLUMN 1          PIC X(9)
                           SOURCE CityName(CityCode).
03  COLUMN 15         PIC 9
                           SOURCE SalesPersonNum.
03  COLUMN 25         PIC $$$,$$.99 SOURCE ValueOfSale.

01  SalesPersonGrp
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.
02  LINE IS PLUS 1.
03  COLUMN 15         PIC X(21)
                           VALUE "Sales for salesperson".
03  COLUMN 37         PIC 9 SOURCE SalesPersonNum.
03  COLUMN 43         PIC X VALUE "=".
03  SMS COLUMN 45     PIC $$$,$$,,$$.99 SUM ValueOfSale.

01  TYPE IS PAGE FOOTING.
02  LINE IS 53.
03  COLUMN 1          PIC X(29)
                           VALUE "Programmer - Michael Coughlan".
03  COLUMN 45         PIC X(6) VALUE "Page :".
03  COLUMN 52         PIC Z9 SOURCE PAGE-COUNTER.

```

REPORT SECTION.

RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.

We want the page heading printed on the same line on each new page so we explicitly state the line number.

01 TYPE IS PAGE HEADING.

02 **LINE 1.**
03 COLUMN 12 PIC X(32)
VALUE "An example COBOL Report Program".

02 **LINE 2.**
03 COLUMN 6 PIC X(17)
VALUE "Bible Salesperson".
03 COLUMN 23 PIC X(26)
VALUE " - Sales and Salary Report".

02 **LINE 4.**
03 COLUMN 2 PIC X(4) VALUE "City".
03 COLUMN 12 PIC X(11) VALUE "Salesperson".
03 COLUMN 28 PIC X(4) VALUE "Sale".

02 **LINE 5.**
03 COLUMN 2 PIC X(4) VALUE "Name".
03 COLUMN 13 PIC X(6) VALUE "Number".
03 COLUMN 28 PIC X(5) VALUE "Value".

01 DetailLine TYPE IS DETAIL.

02 LINE IS PLUS 1.
03 COLUMN 1 PIC X(9)
SOURCE CityName(CityCode).
03 COLUMN 15 PIC 9
SOURCE SalesPersonNum.
03 COLUMN 25 PIC \$\$\$,\$\$.99 SOURCE ValueOfSale.

01 SalesPersonGrp
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.

02 LINE IS PLUS 1.
03 COLUMN 15 PIC X(21)
VALUE "Sales for salesperson".
03 COLUMN 37 PIC 9 SOURCE SalesPersonNum.
03 COLUMN 43 PIC X VALUE "=".
03 SMS COLUMN 45 PIC \$\$\$,\$\$.99 SUM ValueOfSale.

01 TYPE IS PAGE FOOTING.

02 LINE IS 53.
03 COLUMN 1 PIC X(29)
VALUE "Programmer - Michael Coughlan".
03 COLUMN 45 PIC X(6) VALUE "Page :".
03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.

REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.

The COLUMN [IS] phrase allows us to position the characters in each line.

Scroll up to the report page in the previous section to see the result of these phrases.

01 TYPE IS PAGE HEADING.
02 LINE 1.
03 COLUMN 12 PIC X(32)
VALUE "An example COBOL Report Program".

02 LINE 2.
03 COLUMN 6 PIC X(17)
VALUE "Bible Salesperson".
03 COLUMN 23 PIC X(26)
VALUE " - Sales and Salary Report".

02 LINE 4.
03 COLUMN 2 PIC X(4) VALUE "City".
03 COLUMN 12 PIC X(11) VALUE "Salesperson".
03 COLUMN 28 PIC X(4) VALUE "Sale".

02 LINE 5.
03 COLUMN 2 PIC X(4) VALUE "Name".
03 COLUMN 13 PIC X(6) VALUE "Number".
03 COLUMN 28 PIC X(5) VALUE "Value".

01 DetailLine TYPE IS DETAIL.
02 LINE IS PLUS 1.
03 COLUMN 1 PIC X(9)
SOURCE CityName(CityCode).
03 COLUMN 15 PIC 9
SOURCE SalesPersonNum.
03 COLUMN 25 PIC \$\$,\$\$\$\$.99 SOURCE ValueOfSale.

01 SalesPersonGrp
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.
02 LINE IS PLUS 1.
03 COLUMN 15 PIC X(21)
VALUE "Sales for salesperson".
03 COLUMN 37 PIC 9 SOURCE SalesPersonNum.
03 COLUMN 43 PIC X VALUE "=".
03 SMS COLUMN 45 PIC \$\$\$,\$\$\$\$.99 SUM ValueOfSale.

01 TYPE IS PAGE FOOTING.
02 LINE IS 53.
03 COLUMN 1 PIC X(29)
VALUE "Programmer - Michael Coughlan".
03 COLUMN 45 PIC X(6) VALUE "Page :".
03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.

REPORT SECTION.

```
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.
```

01 TYPE IS PAGE HEADING.

```
02 LINE 1.
03 COLUMN 12 PIC X(32)
VALUE "An example COBOL Report Program".
```

02 LINE 2.

```
03 COLUMN 6 PIC X(17)
VALUE "Bible Salesperson".
03 COLUMN 23 PIC X(26)
VALUE " - Sales and Sales"
```

02 LINE 4.

```
03 COLUMN 2 PIC X(4)
03 COLUMN 12 PIC X(11)
03 COLUMN 28 PIC X(4)
```

02 LINE 5.

```
03 COLUMN 2 PIC X(4)
03 COLUMN 13 PIC X(6)
03 COLUMN 28 PIC X(5)
```

Now let's specify the report group required to print the page footing.

As before we must specify the type of report group.
It is a PAGE FOOTING group.

01 DetailLine TYPE IS DETAIL.

```
02 LINE IS PLUS 1.
03 COLUMN 1 PIC X(9)
SOURCE CityName(CityCode).
03 COLUMN 15 PIC 9
SOURCE SalesPersonNum.
03 COLUMN 25 PIC $$$,$$$.$99 SOURCE ValueOfSale.
```

01 SalesPersonGrp

TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.

```
02 LINE IS PLUS 1.
03 COLUMN 15 PIC X(21)
VALUE "Sales for salesperson".
03 COLUMN 37 PIC 9 SOURCE SalesPersonNum.
03 COLUMN 43 PIC X VALUE "="
03 SMS COLUMN 45 PIC $$$,$$$,$$$.$99 SUM ValueOfSale.
```

01 **TYPE IS PAGE FOOTING.**

```
02 LINE IS 53.
03 COLUMN 1 PIC X(29)
VALUE "Programmer - Michael Coughlan".
03 COLUMN 45 PIC X(6) VALUE "Page :".
03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.
```

```
REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.
```

```
01 TYPE IS PAGE HEADING.
02 LINE 1.
03 COLUMN 12 PIC X(32)
VALUE "An ex
```

```
02 LINE 2.
03 COLUMN 6 PIC X(17)
VALUE "Bible Salesperso
03 COLUMN 23 PIC X(26)
VALUE " - Sales and Sal
```

```
02 LINE 4.
03 COLUMN 2 PIC X(4)
03 COLUMN 12 PIC X(11)
03 COLUMN 28 PIC X(4)
```

```
02 LINE 5.
03 COLUMN 2 PIC X(4)
03 COLUMN 13 PIC X(6)
03 COLUMN 28 PIC X(5)
```

```
01 DetailLine TYPE IS DETAIL.
02 LINE IS PLUS 1.
03 COLUMN 1 PIC X(9)
SOURCE Ci
03 COLUMN 15 PIC 9
SOURCE Sa
03 COLUMN 25 PIC $$$,$$
```

```
01 SalesPersonGrp
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP 1 2.
02 LINE IS PLUS 1.
03 COLUMN 15 PIC X(21)
VALUE "Sales for salesperson".
03 COLUMN 37 PIC 9 SOURCE SalesPersonNum.
03 COLUMN 43 PIC X VALUE "="
03 SMS COLUMN 45 PIC $$$,$$,99 SUM ValueOfSale.
```

```
01 TYPE IS PAGE FOOTING.
```

```
02 LINE IS 53.
03 COLUMN 1 PIC X(29)
VALUE "Programmer - Michael Coughlan".
03 COLUMN 45 PIC X(6) VALUE "Page :".
03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.
```

The SOURCE phrase allows a programmer to tell the Report Writer **where** to find the value to be printed.

We want to print a page number in the page footing and we do this by using the SOURCE PAGE-COUNTER phrase.

PAGE-COUNTER is a special register that allows programs to access the page count.

Each report in the Report Section has an associated PAGE-COUNTER


```
REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.
```

```
01 TYPE IS PAGE HEADING.
```

```
02 LINE 1.
03 COLUMN 12 PIC X(32)
VALUE "An
```

```
02 LINE 2.
03 COLUMN 6 PIC X(17)
VALUE "Bible Salespers
03 COLUMN 23 PIC X(26)
VALUE " - Sales and Sa
```

```
02 LINE 4.
03 COLUMN 2 PIC X(4)
03 COLUMN 12 PIC X(11)
03 COLUMN 28 PIC X(4)
```

```
02 LINE 5.
03 COLUMN 2 PIC X(4) VALUE "Name".
03 COLUMN 13 PIC X(6) VALUE "Number".
03 COLUMN 28 PIC X(5) VALUE "Value".
```

The LINE IS PLUS 1 phrase prints the sales information on the next line unless a new page is required.

If a new page is required the line is printed on the next page.

```
01 DetailLine TYPE IS DETAIL.
```

```
02 LINE IS PLUS 1.
03 COLUMN 1 PIC X(9)
SOURCE CityName(CityCode).
03 COLUMN 15 PIC 9
SOURCE SalesPersonNum.
03 COLUMN 25 PIC $$$,$$.99 SOURCE ValueOfSale.
```

```
01 SalesPersonGrp
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.
```

```
02 LINE IS PLUS 1.
03 COLUMN 15 PIC X(21)
VALUE "Sales for salesperson".
03 COLUMN 37 PIC 9 SOURCE SalesPersonNum.
03 COLUMN 43 PIC X VALUE "=" .
03 SMS COLUMN 45 PIC $$$,$$,,$$.99 SUM ValueOfSale.
```

```
01 TYPE IS PAGE FOOTING.
```

```
02 LINE IS 53.
03 COLUMN 1 PIC X(29)
VALUE "Programmer - Michael Coughlan".
03 COLUMN 45 PIC X(6) VALUE "Page :".
03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.
```

REPORT SECTION.
 RD SalesReport
 CONTROLS ARE SalesPersonNum
 PAGE LIMIT IS 66
 HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
 FOOTING 52.

01 TYPE IS PAGE HEADING.
 02 LINE 1.
 03 COLUMN 12 PIC X(32)
 VALUE "An
 02 LINE 2.
 03 COLUMN 6 PIC X(17)
 VALUE "Bible Salespers
 03 COLUMN 23 PIC X(26)
 VALUE " - Sales and Sa
 02 LINE 4.
 03 COLUMN 2 PIC X(4)
 03 COLUMN 12 PIC X(11)
 03 COLUMN 28 PIC X(4)
 02 LINE 5.
 03 COLUMN 2 PIC X(4) VALUE "Name".
 03 COLUMN 13 PIC X(6) VALUE "Number".
 03 COLUMN 28 PIC X(5) VALUE "Value".

PAGE LIMIT specifies the size of the page.

HEADING specifies where the first heading line can be printed.

FOOTING specifies where the first footing line can be printed.

01 DetailLine TYPE IS DETAIL.
 02 LINE IS PLUS 1.
 03 COLUMN 1 PIC X(9)
 SOURCE CityName(CityCode).
 03 COLUMN 15 PIC 9
 SOURCE SalesPersonNum.
 03 COLUMN 25 PIC \$\$,\$\$\$\$.99 SOURCE ValueOfSale.

01 SalesPersonGrp
 TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.
 02 LINE IS PLUS 1.
 03 COLUMN 15 PIC X(21)
 VALUE "Sales for salesperson".
 03 COLUMN 37 PIC 9 SOURCE SalesPersonNum.
 03 COLUMN 43 PIC X VALUE "=".
 03 SMS COLUMN 45 PIC \$\$\$,\$\$\$\$.99 SUM ValueOfSale.

01 TYPE IS PAGE FOOTING.
 02 LINE IS 53.
 03 COLUMN 1 PIC X(29)
 VALUE "Programmer - Michael Coughlan".
 03 COLUMN 45 PIC X(6) VALUE "Page :".
 03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.

```
REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.
```

```
01 TYPE IS PAGE HEADING.
02 LINE 1.
03 COLUMN 12 PIC X(32)
VALUE "An

02 LINE 2.
03 COLUMN 6 PIC X(17)
VALUE "Bible Salespers
03 COLUMN 23 PIC X(26)
VALUE " - Sales and Sa

02 LINE 4.
03 COLUMN 2 PIC X(4)
03 COLUMN 12 PIC X(11)
03 COLUMN 28 PIC X(4)

02 LINE 5.
03 COLUMN 2 PIC X(4)
03 COLUMN 13 PIC X(6)
03 COLUMN 28 PIC X(5)
```

When the salesperson number changes we want to print the total sales for the salesperson.

We can set up a report group (as shown) that describes **what** to print each time the salesperson number changes.

But how will the Report Writer know **when** to print this group?

```
01 DetailLine TYPE IS DETAIL.
02 LINE IS PLUS 1.
03 COLUMN 1 PIC X(9)
SOURCE CityName(CityCode).
03 COLUMN 15 PIC 9
SOURCE SalesPersonNum.
03 COLUMN 25 PIC $$$,$$$$.99 SOURCE ValueOfSale.
```

01 SalesPersonGrp

```
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.
02 LINE IS PLUS 1.
03 COLUMN 15 PIC X(21)
VALUE "Sales for salesperson".
03 COLUMN 37 PIC 9 SOURCE SalesPersonNum.
03 COLUMN 43 PIC X VALUE "=".
03 SMS COLUMN 45 PIC $$$,$$$$.99 SUM ValueOfSale.
```

```
01 TYPE IS PAGE FOOTING.
02 LINE IS 53.
03 COLUMN 1 PIC X(29)
VALUE "Programmer - Michael Coughlan".
03 COLUMN 45 PIC X(6) VALUE "Page :".
03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.
```

```
REPORT SECTION.  
RD SalesReport  
CONTROLS ARE SalesPersonNum  
PAGE LIMIT IS 66  
HEADING 1  
FIRST DETAIL 6  
LAST DETAIL 42  
FOOTING 52.
```

01 TYPE IS PAGE HEADING.
02 LINE 1.
03 COLUMN 12 PIC X(32)
VALUE "An example COBOL Report Program".

02 LINE 2.
03 COLUMN 6 PIC X(17)
VALUE "Bible Salesperson".
03 COLUMN 23 PIC X(26)
VALUE " - Sales and Sa

02 LINE 4.
03 COLUMN 2 PIC X(4)
03 COLUMN 12 PIC X(11)
03 COLUMN 28 PIC X(4)

02 LINE 5.
03 COLUMN 2 PIC X(4)
03 COLUMN 13 PIC X(6)
03 COLUMN 28 PIC X(5)

01 DetailLine TYPE IS DETAIL.
02 LINE IS PLUS 1.
03 COLUMN 1 PIC X(9)
SOURCE C
03 COLUMN 15 PIC 9
SOURCE Sa
03 COLUMN 25 PIC \$\$\$,\$\$\$.

01 **SalesPersonGrp**
TYPE IS CONTROL FOOTING **SalesPersonNum** **NEXT GROUP PLUS 2.**
02 LINE IS PLUS 1.
03 COLUMN 15 PIC X(21)
VALUE "Sales for salesperson".
03 COLUMN 37 PIC 9 SOURCE SalesPersonNum.
03 COLUMN 43 PIC X VALUE "=".
03 SMS COLUMN 45 PIC \$\$\$,\$\$\$,\$\$\$.\$99 SUM ValueOfSale.

01 TYPE IS PAGE FOOTING.
02 LINE IS 53.
03 COLUMN 1 PIC X(29)
VALUE "Programmer - Michael Coughlan".
03 COLUMN 45 PIC X(6) VALUE "Page :".
03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.

When we specify that a group is a control group, we must also specify the name of the control break data-item. The Report Writer monitors the value in this item and when it changes a control break occurs.

But if there is more than one control break item how does the Report Writer establish the relationship between them?

```
REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.
```

```
01 TYPE IS PAGE HEADING.
02 LINE 1.
03 COLUMN 12 PIC X(32)
VALUE "An example COBOL Report Program".

02 LINE 2.
03 COLUMN 6 PIC X(17)
VALUE "Bible Salesperson".
03 COLUMN 23 PIC X(26)
VALUE " - Sales and Sa

02 LINE 4.
03 COLUMN 2 PIC X(4)
03 COLUMN 12 PIC X(11)
03 COLUMN 28 PIC X(4)

02 LINE 5.
03 COLUMN 2 PIC X(4)
03 COLUMN 13 PIC X(6)
03 COLUMN 28 PIC X(5)

01 DetailLine TYPE IS DETAIL.
02 LINE IS PLUS 1.
03 COLUMN 1 PIC X(9)
SOURCE CityName(CityCode).
03 COLUMN 15 PIC 9
SOURCE SalesPersonNum.
03 COLUMN 25 PIC $$$,$$.99 SOURCE ValueOfSale.
```

The order of the control breaks is established by the **CONTROLS ARE** phrase in the report's RD entry. The control break items are given in order of seniority. If a senior control break occurs, it automatically causes a control break on all items junior to it.

```
01 SalesPersonGrp
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.
02 LINE IS PLUS 1.
03 COLUMN 15 PIC X(21)
VALUE "Sales for salesperson".
03 COLUMN 37 PIC 9 SOURCE SalesPersonNum.
03 COLUMN 43 PIC X VALUE "="
03 SMS COLUMN 45 PIC $$$,$$.99 SUM ValueOfSale.

01 TYPE IS PAGE FOOTING.
02 LINE IS 53.
03 COLUMN 1 PIC X(29)
VALUE "Programmer - Michael Coughlan".
03 COLUMN 45 PIC X(6) VALUE "Page :".
03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.
```

```
REPORT SECTION.  
RD SalesReport  
CONTROLS ARE SalesPersonNum  
PAGE LIMIT IS 66  
HEADING 1  
FIRST DETAIL 6  
LAST DETAIL 42  
FOOTING 52.  
  
01 TYPE IS PAGE HEADING.  
02 LINE 1.  
03 COLUMN 12 PIC X(32)  
VALUE "An example COBOL Report Program".  
  
02 LINE 2.  
03 COLUMN 6 PIC X(17)  
VALUE "Bible Salesperson".  
03 COLUMN 23 PIC X(26)  
VALUE " - Sales and Salesperson".  
  
02 LINE 4.  
03 COLUMN 2 PIC X(4)  
03 COLUMN 12 PIC X(11)  
03 COLUMN 28 PIC X(4)  
  
02 LINE 5.  
03 COLUMN 2 PIC X(4)  
03 COLUMN 13 PIC X(6)  
03 COLUMN 28 PIC X(5)  
  
01 DetailLine TYPE IS DETAIL.  
02 LINE IS PLUS 1.  
03 COLUMN 1 PIC X(9)  
SOURCE CityName(CityCode).  
03 COLUMN 15 PIC 9  
SOURCE SalesPersonNum.  
03 COLUMN 25 PIC $$$,$$.99 SOURCE ValueOfSale.  
  
01 SalesPersonGrp  
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.  
02 LINE IS PLUS 1.  
03 COLUMN 15 PIC X(21)  
VALUE "Sales for salesperson".  
03 COLUMN 37 PIC 9 SOURCE SalesPersonNum.  
03 COLUMN 43 PIC X VALUE "=".  
03 SMS COLUMN 45 PIC $$$,$$.99 SUM ValueOfSale.  
  
01 TYPE IS PAGE FOOTING.  
02 LINE IS 53.  
03 COLUMN 1 PIC X(29)  
VALUE "Programmer - Michael Coughlan".  
03 COLUMN 45 PIC X(6) VALUE "Page :".  
03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.
```

We have identified the item that contains the salesperson total with the name SMS.

In this program we don't actually need to identify this item because we never refer to it but it is declared now because we will refer to it when we extend the program.

PROCEDURE DIVISION.

Begin.

OPEN INPUT SalesFile.

OPEN OUTPUT PrintFile.

READ SalesFile

AT END SET EndOfFile TO TRUE

END-READ.

INITIATE SalesReport.

PERFORM PrintSalaryReport

UNTIL EndOfFile.

TERMINATE SalesReport.

CLOSE SalesFile, PrintFile.

STOP RUN.

PrintSalaryReport.

GENERATE DetailLine.

READ SalesFile

AT END SET EndOfFile TO TRUE

END-READ.

The INITIATE statement sets up the report and in it we specify which report we want to set up.

The TERMINATE closes down the report and we must specify which report to close.

The GENERATE statement prints the detail lines and must specify the name we have assigned to the DETAIL group.

Each time we read a record we generate a detail line.

End of commentary